

Joseph Kerkhof

☎ +1 920 376 1248 | ✉ joseph@kerkhof.dev | 📍 New London, WI
🌐 kerkhof.dev | 🐙 github.com/josephkerkhof | in linkedin.com/in/josephkerkhof

SUMMARY

Senior Backend Engineer with 7+ years building and running production systems, mostly in PHP, plus Python and Go. Owns work from design through operations: data models, queues, integrations, CI/CD, monitoring, and upgrades. Recently led an AI-assisted email production pipeline to launch and built a company-wide OpenTelemetry platform.

TECHNICAL SKILLS

Backend & Data: PHP, Laravel, Go, Python, PostgreSQL, MySQL/MariaDB, Redis, Snowflake, REST APIs

AI Engineering: Laravel AI, LangGraph, Claude Code, OpenCode, MCP Integrations, Agent Orchestration

Cloud & Delivery: AWS, Cloudflare, Linux, Docker, Kubernetes, GitHub Actions, CI/CD

Architecture & Quality: DDD, CQRS, Microservices, Pest, PHPUnit, PHPStan, Pint, TDD, Load Testing

Observability & Identity: OpenTelemetry, ClickStack/HyperDX, Distributed Tracing, Enterprise SSO

WORK EXPERIENCE

Active Engagement

Remote, USA

Senior Developer

Mar 2026 – Present

Tech Stack: Laravel, Laravel AI, PostgreSQL, Python, Kubernetes, OpenTelemetry

- Led an AI-assisted email production pipeline from plan to launch in four weeks: client-approved copy passes a QA gate, then becomes CRM records and an email draft automatically, replacing manual build steps. Live for client emails since Sep 2026.
- Made an approval-to-CRM job chain safe to retry: approvals are frozen as content-hashed snapshots, each external write is a checkpointed step, and unclear CRM responses park the run for reconciliation instead of creating duplicates.
- Built the company's first AI agent, which turns a creative brief into an email draft, in Python/LangGraph with structured LLM outputs and deterministic fallbacks; later moved it into the main Laravel app behind a parity-checked dark launch and retired the separate service.
- Started and led a company-wide observability platform on self-hosted ClickStack, OpenTelemetry, and Kubernetes instead of a paid vendor; it now covers 7 apps with tracing, alerts, and session replay. Used it to find production users on an unmonitored staging backend and moved them in one afternoon.
- Own production operations across bare-metal and cloud Linux servers: led three MySQL 8.0 to 8.4 cluster upgrades before end of life (catching a schema break first), patched a server 435 packages behind in a 35-minute window, and added nightly backups with a tested restore.
- Restored a static-analysis gate that a merge had silently disabled for a month: PHPStan at max level with no baseline, about 550 findings cleared, two production bugs fixed. Cut median CI time about 40% with timing-balanced test shards.
- Introduced PR reviews and CI checks to a newly grown dev team, and mentored junior engineers through pairing and code review.

Renewed Vision

Remote, USA

Senior Laravel Developer

Dec 2025 – Feb 2026

Tech Stack: Laravel, PHP

Active Engagement

Remote, USA

Senior Developer (Contract)

Oct 2025 – Nov 2025

Tech Stack: Laravel, Horizon, Snowflake, MySQL

- Built the Snowflake-to-Laravel data pipeline behind a new client reporting platform: an hourly dispatcher, concurrency-capped Horizon queues, and batched upserts; a 15-month backfill ran in about 4 minutes.

Teamworks

Senior Software Engineer

Remote, USA

May 2024 – Sep 2025

Tech Stack: Laravel, PHP, MySQL, Redis, AWS, Kubernetes, CI/CD

- Architected and delivered GM budget management product from the ground up in 3 months, processing athlete compensation across multiple universities with 90% test coverage and daily production deployments via automated CI/CD pipelines.
- Delivered integrated payment systems in 8 months ahead of the NCAA's July 2025 regulatory deadline, using Domain-Driven Design, CQRS, and action-based architecture to keep complex regulatory logic maintainable and testable.
- Engineered a high-performance RESTful API achieving sub-second response times through optimized database queries and asynchronous processing with Redis queues, implementing comprehensive observability via structured logging, distributed tracing, and StatsD metrics.

Prodege

PHP Developer

Remote, USA

Mar 2023 – May 2024

Tech Stack: PHP, Laravel, MySQL, Google Ads SDK, Cron Jobs

- Built an automated Google Ads campaign management system that cut launch time from 8 hours to 30 minutes and prevented \$20K–\$30K in monthly wasted spend through real-time budget optimization and alerting.
- Modernized legacy PHP cron job infrastructure to Laravel 10, eliminating 15+ monthly job failures and reducing system maintenance time through improved error handling and monitoring.
- Optimized MySQL queries on 200GB+ tables, reducing query execution time from 45 seconds to under 3 seconds and improving dashboard load times by 85% through table redesign and proper indexing strategies.
- Identified and remediated critical vulnerabilities including remote code execution, SQL injection, and exposed credentials.

Clockwork

Senior Software Engineer

Minneapolis, MN

Mar 2019 – Mar 2023

Tech Stack: PHP, MySQL, AWS Serverless, React, TypeScript, Jest, Load Testing Tools

- Built web products serving 50K+ users as team Delivery Lead, managing sprint planning for 6-person team while architecting React/TypeScript frontends.
- Load-tested high-traffic website processing 100K+ daily requests to identify application and infrastructure bottlenecks, implementing horizontal scaling solutions that improved response times by 60% and eliminated timeout errors.
- Maintained a PHP/MySQL promotion engine serving 350K+ accounts and thousands of daily transactions with 99.9% uptime; participated in on-call support for Minnesota Lottery applications.

EDUCATION

University of Wisconsin — Oshkosh

Oshkosh, WI

M.Sc. in Data Science (Part-time); GPA: 4.00/4.00

Sep 2017 – Jun 2021

B.Sc. in Computer Science

Sep 2011 – Jun 2016

PROJECTS

Omnibus Config File Validator | [GitLab](#)

Tech Stack: Golang, Cobra

- Built a Go CLI that validates GitLab Omnibus configurations for security issues and common misconfigurations, with automated testing, linting, and builds in GitLab CI.

GitHub RSS | [GitHub](#)

Tech Stack: PHP, Laravel, Inertia, Vue.js

- Built a GitHub activity feed service around interface-driven dependencies and in-memory test doubles, enabling fast, isolated unit tests while supporting REST APIs with limit-offset and cursor pagination.